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Yolo-HLSAM: Adapting foundation segment anything model for semi-automatic detection and segmentation of breast cancer microcalcification clusters

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ABSTRACT

Microcalcification segmentation plays an important role in the detection and diagnosis of breast cancer. While many deep neural network models perform well on standard datasets, their generalizability is often limited due to variations in data sources, device types, and other factors, leading to reduced accuracy in practical clinical applications. The Segment Anything Model (SAM), known for its zero-shot generalization capability, has recently gained significant attention in natural image segmentation tasks. Although the enhanced High-Quality SAM (HQ-SAM) model has achieved remarkable success in improving segmentation accuracy for ambiguous boundaries, it still struggles with precisely delineating small targets like microcalcifications. Therefore, this study proposes a Yolo-HLSAM-based approach for segmenting breast cancer microcalcification clusters. By integrating the precision of Yolov7 with the advanced segmentation capabilities of the HLSAM model, our method not only ensures accurate lesion detection but also significantly improves segmentation precision, while maintaining the scalability of the original SAM architecture. Additionally, a semi-automatic annotation technique is incorporated into the Yolo-HLSAM framework to further improve segmentation performance. Experimental results demonstrate that our approach substantially outperforms existing methods in segmentation accuracy, with Dice coefficients of 89.19 ±0.26% and 88.76 ±0.32% for the private and VinDr-Mammo datasets, respectively. Our code is available on https://github.com/ccchg/yolo-hlsam.

1. Introduction

As technological advancements accelerate, deep learning-based medical image segmentation techniques have become central to the development of efficient computer-aided diagnosis (CAD) systems, attracting significant attention and recognition in the field of medical imaging. These techniques offer remarkable precision in delineating lesion areas, greatly assisting physicians in making accurate diagnoses and formulating treatment strategies. This enhances the efficiency and accuracy of disease diagnosis and treatment [1]. Medical image segmentation has found increasing application in breast cancer imaging analysis [2–4]. By labeling each pixel and classifying it into a specific category, these methods identify regions of interest within the original images, establishing segmentation as one of the most prominent research areas in both computer vision and medical image analysis [5]. Accurate segmentation provides reliable information about the volume and shape of target structures, further facilitating clinical applications [6,7]. As evidenced by relevant research, its accuracy has even surpassed human experts in certain domains.

Conventional methodologies for breast cancer segmentation, such as threshold method and edge detection [8–10], are highly sensitive to image contrast. The segmentation accuracy of these approaches may be compromised when the contrast between the breast cancer lesion and the surrounding tissue is low. In addition, these methodologies show undesirable adaptability to complex lesions, making it difficult to accurately segment lesions that are irregularly shaped or have blurred boundaries. This limitation is tremendously conspicuous in the early stages of breast cancer, where lesions are often small and inconspicuous, which thereby gives rise to sub-optimal segmentation results with conventional methodologies [11]. As a result, a multitude of researchers have incorporated mainstream deep learning methodologies,

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such as Convolutional Neural Networks (CNNs) and U-Net, into the field of medical image segmentation. It is particularly noteworthy that CNNs can automatically learn and extract image features, capturing hierarchical features from simple to complex across multiple layers, thereby elevating the understanding of image content [12–14].

In deep neural networks, continuous pooling lowers spatial resolution, thus bring about loss of location information. Simultaneously, despite the fact that deep learning methodologies (such as U-Net, Mask R-CNN, etc.) have accomplished substantial advancements in breast cancer image segmentation, boundary ambiguity still exists in the segmentation results when processing subtle lesions, calcification points and other areas. Worse still, detail loss is tremendously prominent especially when the contrast between the lesion and the surrounding tissue is low. The segmentation accuracy is easily affected on account of this adverse aspect. Apart from that, numerous models perform well on standard datasets, but lack the ability to generalize as a consequence of differences in data sources, device types, scanning parameters, etc., which thereby results in a decline in their accuracy in practical clinical applications. To perfectly deal with the limitations of these methodologies in segmentation performance, the SAM was incorporated correspondingly, which can be attributd to its outstanding zero-shot generalization capabilities and broad application potential in natural image segmentation [15]. But the basic SAM model was evaluated on common medical datasets such as FLARE 2022. The results indicated that the model performed well on targets with clear boundaries but struggled with targets that have indistinct boundaries and continuous structures [16].

Overall, although the SAM model has excellent generalization ability, it still faces two challenges when processing highly complex medical images: high manual labeling costs and suboptimal segmentation accuracy. To address these issues, this paper proposes a revolutionary Yolo-HLSAM model, which combines the improved Yolov7 neural network with the HLSAM segmentation framework. Firstly, Yolov7 is used to accurately detect potential lesion areas in FFDM images and output candidate boxes to reduce interference from irrelevant areas. Next, a brightness threshold extraction algorithm is used to isolate prominent highlights within these regions. These isolated highlights are then fed into the HLSAM model for precise focal segmentation. Compared with the traditional SAM model, HLSAM significantly enhances the ability to capture local details and integrate global information through the fusion of multi-scale features. In addition, the segmentation accuracy is further optimized by introducing the boundary sensing loss function, resulting in more refined masks and clearer boundaries. This Yolo-HLSAM model not only optimizes the segmentation accuracy but also significantly reduces the need for manual intervention. The major contributions of this paper are summarized as follows:

(1) Introduction of attention mechanism: CBAM attention mechanism was added to the Yolov7 network, which enables it to fix its attention on important areas and improve accuracy.

(2) Eliminate the model's dependence on manual prompts: Through the region box generated by Yolov7, the point with the highest brightness is automatically extracted as the input of HLSAM model, thus avoiding the dependence on manual prompts.

(3) Introduction of boundary perception loss function: Design and introduce boundary perception loss function to augment the accuracy of the model to the lesion boundary by increasing the weight of the boundary region.

2. Related work

2.1. Deep learning-based medical image segmentation

In recent years, the application of deep learning technology has achieved remarkable successes in the field of medical imaging, particularly in breast cancer detection. Methodologies like YOLO (You Only Look Once) have received extensive attention both at home and abroad. Thanks to its efficient real-time target detection capabilities, YOLO is especially suitable for the rapid detection of tumors and lesions in medical images. Baccouche et al. [17] conducted an in-depth exploration of the effectiveness of the end-to-end fusion model based on the YOLO architecture in simultaneously detecting and classifying suspicious breast lesions on digital mammography. Their aim was to enhance the accuracy of early lesion identification through deep learning technology, reduce unnecessary biopsies, and decrease mortality. In the UCHCDMdatabase data set (private dataset), the result of Acc is 90% ± 0.06.

In addition, Su et al. [18] integrated a dual model of YOLO and LOGO (Local-Global) architecture for the detection and segmentation of breast cancer lumps. Initially, the YOLOv5L6 model was employed to locate and clip the tumor in the mammogram. Subsequently, the LOGO training strategy was used to train the global and local converter branches on the full image and the cropped image respectively. Finally, the two branches were combined for the final segmentation decision.

Meanwhile, Prinzi et al. [19] developed an automated breast cancer detection system based on the YOLO model to assist physicians in making decisions during breast cancer screening. Specifically, significance maps were generated by applying Eigen-CAM technology to help the model interpret the predictions and highlight suspicious areas in the mammograms.

Concurrently, some researchers have focused on the application of U-Net and its variants, which not only enhance segmentation accuracy and robustness through multi-scale feature fusion but also improve network architectures. For example, Vida et al. [20] proposed a U-Net ensemble method that effectively captures the complex morphology of breast lesions in DCE-MRI by combining multiple U-Net models and multi-scale feature extraction. This method is especially proficient in handling irregular morphologies.

Phuong et al. [21] utilized the U-Net architecture to segment abnormal regions in breast cancer X-ray images, significantly improving the accuracy of automated segmentation and thereby facilitating early detection. Pramanik et al. [22] devised the DAU-Net model, which incorporates spatial and channel attention mechanisms, greatly enhancing tumor segmentation accuracy in ultrasound images with noise and complex backgrounds. It is worth emphasizing that these studies have jointly promoted technological advancements in breast cancer detection, thereby enhancing the reliability of clinical diagnoses.

YOLO models perform outstandingly in target detection tasks, featuring high detection speed and accuracy, and are suitable for real-time applications. However, they require a large amount of high-quality labeled data, which makes the labeling process complex and time-consuming. Moreover, they are also sensitive to image noise and low-quality images, which may impact the detection results.

2.2. SAM model-based image segmentation

Mainstream neural network architectures, such as GANs (Generative Adversarial Networks) and U-Net, are typically optimized for specific datasets. However, this fine-tuning process can limit their generalization capabilities. More concerning is the potential for significant performance degradation when these models are applied to scenarios that differ from the original training data, ultimately restricting their range of applicability. To overcome these limitations, Meta introduced the SAM, a versatile large-scale model engineered to execute one-click segmentation on any object present in photos or videos. This innovation improves the model's adaptability and practical utility across a diverse array of scenarios. Notably, some researchers have evaluated SAM's base model on medical images to explore its potential applications in the medical domain. These investigations seek to gauge SAM's efficacy in dealing with intricate medical images and to ascertain its aptness and precision for medical image segmentation tasks.

Mazurowski et al. [23] investigated the application of SAM in medical image analysis, conducting experiments to assess SAM's performance with medical image data. Their analysis took into consideration

Table 1
Related work literature review summary.

Document	Method	Data set	Quantitative result
[17]	YOLO + Image to image conversion	UCHCDM database (private dataset)	Acc=90% ± 0.06
[18]	YOLO-LOGO	DDSM	Dice=0.74, IoU=64%
[19]	YOLOv5	DDSM	Acc=89%
[20]	U-Net Integrated approach	DCE-MRI	IoU = 0.85
[21]	U-Net	INbreast, MIAS	IoU = 0.79, Dice = 0.85
[22]	DAU-Net	BUSI,UDIAT	IoU = 0.82, Dice = 0.88
[23]	SAM	MIAS	Dice ≈ 0.70
[24]	Med-SA	MIAS	Dice>0.85
[25]	SAMed	MICCAI 2015	Dice = 0.82
[26]	HQ-SAM	DIS, ThinObject-5K, COIFT and HR-SOD	Dice ≈ 0.85

the model's applicability and limitations, confirming SAM's potential in medical image segmentation and healthcare domain. However, in dataset MIAS, the results achieved are not ideal, Dice is only 0.7.

Wu et al. [24] suggested the Med-SA model, which incorporated the Spatial-Depth Transpose (SD-Trans) and Hyper Prompt Adapter (HyP-Adpt) mechanisms. This approach maintained the majority of SAM's pre-trained parameters in a frozen state, enabling efficient segmentation of medical images. In dataset MIAS, Dice is increased by 15% compared to SAM. Zhang et al. [25] innovated SAMed, an optimization of SAM tailored for medical image segmentation. They incorporated a Low-Rank Adaptation (LoRA) fine-tuning strategy, which allowed SAMed to efficiently process medical images. On top of that, training strategies, such as warmup and the AdamW optimizer, were employed successively, this strikingly heightening the model's convergence speed and segmentation accuracy.

Ke et al. [26] incorporated HQ-SAM, a model that retains the zero-shot segmentation capability of the original SAM model while remarkably heightening segmentation accuracy through minimal adjustments. By incorporating High-Quality Output Tokens (HQ-Output Tokens) and a Global–Local Feature Fusion mechanism, HQ-SAM generates more refined segmentation boundaries and excels in handling complex object structures. Nonetheless, the model continues to exhibit deficiencies when confronted with small target samples.

These studies collectively underscore the potential of SAM and its derivatives in medical image segmentation, offering promising directions for future research and application in this critical field. Table 1 systematically summarizes a literature review of associated work.

3. A SAM-based method for segmenting breast microcalcification clusters

Grounded in Yolo-HLSAM, the novel segmentation paradigm for breast microcalcification clusters recommended in this study, primarily consists of three components, namely the Yolov7-based region proposal network, image segmentation by employing the HLSAM model, and semi-automatic annotation via Yolo-HLSAM. The overall framework of this approach is illustrated in Fig. 1.

3.1. Yolov7-based region proposal network

To lessen the manual annotation cost associated with constructing prompt engineering, the suggested Yolo-HLSAM method utilizes Yolov7 to extract regions of interest (ROIs) and selects high-intensity spots within these regions as point prompts.

In the context of medical diagnostic image processing, Yolov7 is employed to delineate specific lesion areas and to extract a high-intensity spot from each lesion area as input for the HLSAM model. What deserves mention is that this approach principally addresses two pivotal objectives. First, this methodology is intended to accurately identify and localize lesion areas within medical images, thereby heightening diagnostic accuracy. Aside from that, it aims to swiftly recognize lesion areas and use them as inputs for the SAM model, thereby lessening unnecessary computational load and elevating overall segmentation efficiency.

To reinforce Yolov7's sensitivity to small objects, the Mosaic data augmentation method is incorporated [27–29]. The FFDM suspicious region localization algorithm on the basis of the Yolov7 convolutional neural network constructs a dataset by utilizing FFDM data and employs the Yolov7 network to detect lesions. The Yolov7 network performs feature extraction by employing convolutional layers and pooling layers, and it optimizes target localization by constructing a Complete Intersection over Union (CIOU) loss function. The specific implementation process is as follows:

When constructing the FFDM Dataset, it is essential to define the total number of patients as N. Aside from that, each patient is associated with four FFDM images, which are acquired from distinct positions. The dataset is created by treating individual patients as the smallest unit, thereby triggering image datasets and corresponding label datasets.

$$I = \left(m_1, m_2, m_3, \ldots, m_N\right), \tag{1}$$

$$L = \left(l_1, l_2, l_3, \ldots, l_N\right), \tag{2}$$

Where, m represents the individual image data, and l refers to the corresponding label data.

The architecture of the Yolov7 convolutional neural network is illustrated in Fig. 2. Given that the microcalcification cluster lesions are small targets and occupy a comparatively smaller portion of the image in contrast to the background, the model tends to be biased toward predicting the larger background regions. To cope well with this issue, the CIOU loss function is employed accordingly. The CIOU loss is advantageous to correct the imbalance in matching positive and negative samples, thus augmenting the model's concentration on the smaller lesion areas. The CIOU loss function is defined as follows:

$$L_{CIOU} = 1 - IOU + \frac{\rho\left(b, b^{gt}\right)}{c^2} + av, \tag{3}$$

Where, IOU representing the intersection and union ratio of the predicted box and the real box. b and b^{gt} represent the center points of the predicted and true bounding boxes, severally, $\rho\left(b, b^{gt}\right)$ defines the Euclidean distance, c refers to the diagonal distance of the minimum closure region that can contain both the predicted and true boxes, a is an equilibrium coefficient and v denotes the consistency of the aspect ratio between the predicted bounding box and the real bounding box.

To strengthen the representational and generalization capabilities of the Yolov7 network, the Convolutional Block Attention Module (CBAM) [30,31] is incorporated accordingly. CBAM improves model performance by integrating channel and spatial attention mechanisms

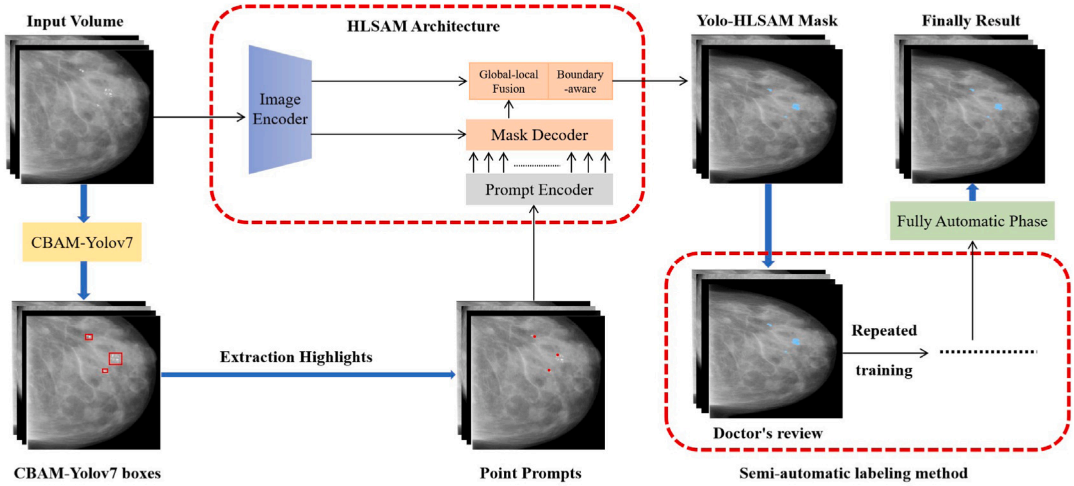

Fig. 1. Flow diagram of breast cancer microcalcification cluster segmentation method on the basis of Yolo-HLSAM.

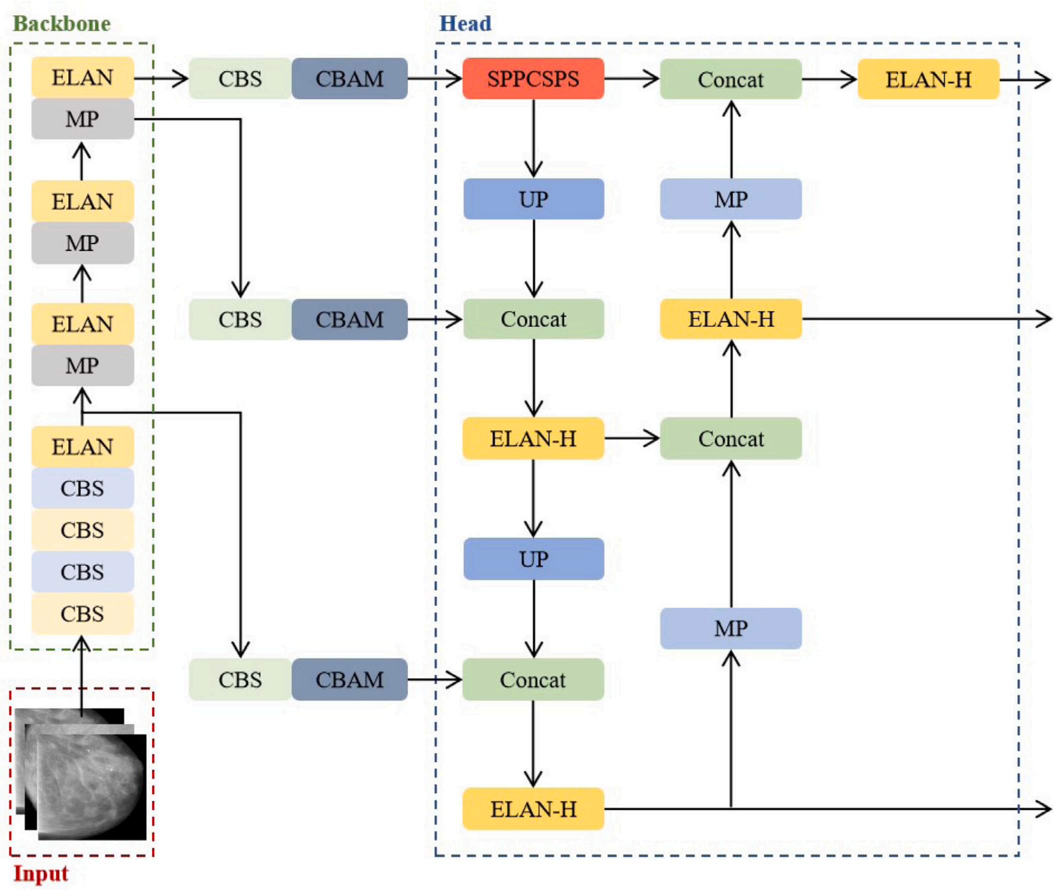

Fig. 2. Add CBAM to Yolov7 network structure.

within the convolutional neural network, thereby enabling the network to concentrate on more informative features and ignore less relevant ones. The detailed architecture incorporating CBAM is depicted in Fig. 2.

CBAM (Convolutional Block Attention Module) consists of two pivotal modules, namely, the Channel Attention Module and the Spatial Attention Module. Specifically, the Channel Attention Module allocates weights to the feature maps across dissimilar channels by concentrating on the attention regions between channels, thereby elevating important features and suppressing irrelevant ones. By comparison, the Spatial Attention Module, predominantly center around attention regions within a single-channel image, which is advantageous for calculating spatial attention weights to highlight significant spatial positions. By integrating these two modules, CBAM effectively augments the model's ability to perceive lesions across diverse channels and spatial locations. The detailed operations of CBAM are described as follows:

Channel Attention Mechanism: Given an input feature map $F \in R^{C\times H\times W}$ where C denotes the number of channels, which H and

W represent the height and width, severally. The channel attention mechanism is applied as follows: To start with, global average pooling is computed in line with Eq. (4). Subsequently, a fully connected layer is applied as described in Eq. (5). At last, the channel attention weights are generated, as presented in Eq. (6). These weights are subsequently multiplied with each channel of the input feature map, thereby elevating or suppressing the feature representation of each channel.

$$F_{avg} = \frac{1}{H \times W} \sum_{i=1}^{H} \sum_{j=1}^{W} F_c(i, j), \tag{4}$$

$$F_{FC} = \mathrm{Re}LU\left(W_2 \delta\left(W_1 F_{avg}\right)\right), \tag{5}$$

$$W_C = sigmoid(F_{FC}), \tag{6}$$

Where, $F_c(I, J)$ refers to the elements of all channels at the i row and j column of the feature map. W_1 and W_2 are learnable weight matrices. While δ symbolizes the activation functions.

Spatial Attention Mechanism: For the input feature map $F \in R^{C \times H \times W}$, spatial attention is applied as follows: Above all, both max-pooling and average-pooling are computed along the channel dimension. The results of these two pooling operations are subsequently concatenated along the channel axis, as described in Eq. (7). Afterwards, a convolution operation followed by an activation function is applied to generate the spatial attention weights, as demonstrated in Eq. (8). In the end, these spatial attention weights are multiplied by the input feature map, as indicated in Eq. (9), thereby elevating or suppressing the feature representation at specific spatial locations.

$$F_{concat} = concat(F_{\max} F_{avg}), \tag{7}$$

$$W_S = sigmoid(W_3 \delta(W_4 F_{concat})), \tag{8}$$

$$X_{att} = W_S \cdot X, \tag{9}$$

Where, $concat$ stands for concatenation. $F_{\max}$ and F_{avg} denote the results of max-pooling and average-pooling, separately. W_3 and W_4 are learnable weight matrices.

CBAM Attention Feature Map: By multiplying the channel attention feature map with the spatial attention feature map, the feature representation is refined, allowing the model to fix its attention on the most informative features in both dimensions. This approach strengthens YOLOv7's ability to detect calcified lesions in breast cancer. The final output of CBAM is:

$$X_{out} = X_{att} \times W_C, \tag{10}$$

4. Image segmentation network grounded in the HLSAM model

To deal well with the issues of segmentation errors and omissions in small target samples within medical images through the utilization of the HQ-SAM model, this paper come up with an optimized segmentation algorithm, termed HLSAM. This algorithm integrates a boundary-aware loss function to generate highly accurate image masks, thereby establishing a robust foundation for semi-automatic annotation. As illustrated in Fig. 3.

In the HQ-SAM model, the incorporation of the boundary-aware loss function fundamentally refines the training process by augmenting the weighting of boundary regions, thereby heightening the model's concentration on the precision of these critical areas. The use of this loss function effectively resolves around the following two pivotal issues:

Precise Segmentation of Object Boundaries: The boundary-aware loss function ameliorates the model's ability to identify and accurately segment the boundary regions of targets by generating boundary masks. These boundary masks are customarily derived from ground truth segmentation masks through the application of sophisticated edge detection algorithms, and subsequently, they are contrasted with the predicted masks during the training phase. This approach enables the model to better deal with complex boundaries, thereby heightening segmentation accuracy.

Elevated Accuracy in Small Object Segmentation: Since small objects typically occupy only a minimal pixel area in images, conventional global loss functions might overlook the precise segmentation of these regions. The boundary-aware loss function, nevertheless, emphasizes boundary regions, ensuring that the model maintains high accuracy when dealing with small targets. On this basis, this approach has invited a remarkable optimization in the model's performance on small object segmentation tasks.

When the boundary-aware loss function is incorporated into the HLSAM model, the function is incorporated subsequent to the global feature fusion step. At this stage, a boundary mask is generated, which explicitly identifies the regions in the image that require focused attention—namely, the boundaries of the target objects.

The implementation of the boundary-aware loss function considerably rests with the accurate identification and extraction of boundary regions from the ground truth segmentation masks. To extract these boundary regions, an edge detection algorithm is employed to generate the boundary mask. In detail, the boundary mask is a binary image where pixels labeled as "1" represent boundary regions, while pixels labeled as "0" represent non-boundary regions. The Sobel operator is utilized to calculate the gradient changes in both the horizontal and vertical directions of the image. FRegions with higher gradient values are marked as boundaries. This approach effectively captures the boundary information of target objects, which enables the boundary mask to represent the essential edge features.

Upon the generation of the boundary mask, it is combined with the model's predicted segmentation mask for loss calculation. To further strengthen the model's concentration on boundary regions, a dilation operation is applied to the boundary regions, expanding their influence. This expansion facilitates the model in more precisely processing single-pixel boundaries. Furthermore, it enhances the model's capability to capture and optimize the comprehensive boundary features within the adjacent vicinity. Consequently, both segmentation accuracy and robustness are significantly augmented.

To clearly express this concept, the boundary-aware loss function is defined as follows: by assigning higher weights to the pixels in the boundary regions, the model is encouraged to concentrate on the segmentation accuracy of these areas during training. The boundary-aware loss function can be represented as:

$$L_{boundary} = \frac{1}{|B|} \sum_{x \in B} Loss(P(x), G(x)), \tag{11}$$

Where, B denotes the set of pixels in the boundary region, $P(x)$ refers to the segmentation mask value predicted by the model, $G(x)$ is the true segmentation mask value, and $Loss$ defines the cross-entropy loss.

The total loss function combines the boundary-aware loss function with the global loss function to balance overall segmentation accuracy with boundary precision:

$$L_{total} = \alpha \cdot L_{global} + \beta \cdot L_{boundary}, \tag{12}$$

Where, L_{global} symbolizes the original global loss (cross-entropy loss), α and β are weight parameters that regulate the contribution of the global loss and margin loss to the total loss.

This combined loss approach ensures that the model maintains a balance between materializing high overall segmentation accuracy and focusing on the precision of object boundaries, thus addressing the challenges of segmenting small or complex objects more effectively.

5. Semi-automatic annotation grounded in Yolo-HLSAM model

To augment the efficiency of data annotation and model training in medical image segmentation tasks, and to alleviate the high demands

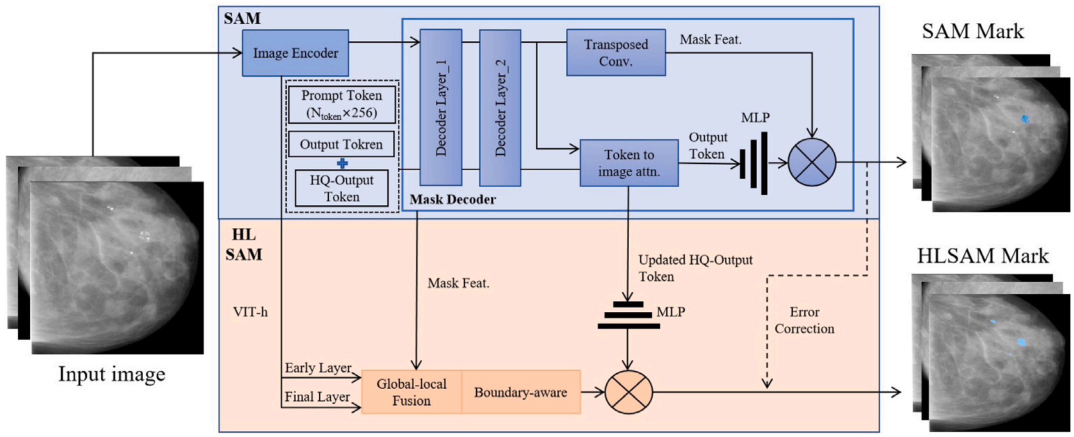

Fig. 3. Segmentation flow chart on the basis of HLSAM model.

on human resources and time in conventional data annotation processes [2,32], a semi-automatic annotation approach is incorporated on the basis of Yolo-HLSAM. This method combines labeled and unlabeled data to strengthen model performance, elevate overall research efficiency, and lessen the annotation cost associated with Yolov7. The entire semi-automatic annotation process can be divided into three phases:

5.1. Assisted manual phase

In the first phase, the Yolo-HLSAM model is employed to perform a preliminary analysis and processing of Full Field Digital Mammography (FFDM) images. The model detects potential breast cancer microcalcification cluster regions within the images and generates initial annotation masks. These segmentation results are typically presented as binary masks, where the pixel values within the array are set to "1" to indicate pixels that require further processing (i.e., potential lesions), while values of "0" represent non-lesion background.

To enhance mask accuracy, particularly around irregularly shaped boundary pixels, the process involves extracting the X and Y coordinates of these boundary pixels. On this basis, the minimal bounding rectangle enclosing the shape is determined using a min–max update mechanism, which optimizes the shape boundaries. This approach improves the precision of the masks. The coordinates of all non-zero pixels in the masks are expressed as follows:

$$S = \{(x, y) \,|mask\,[y]\,[x] = 1\,\}, \tag{13}$$

S denotes a set containing pairs of 2D coordinates (x, y), where (x, y) refers to a 2D coordinate point, and $mask[y][x]$ symbolizes the pixel value at row y and column x in the mask.

5.2. Semi-automatic phase

In the semi-automatic phase, pre-trained models are deployed to automate the initial annotation process, thereby bettering the performance of annotation on FFDM images. The annotations generated during this process are reviewed by medical experts. It is crucial to point out that this expert review not only corrects any potential errors originated from the model but also fixes attention on specific lesion areas that the model may have failed to segment accurately. This step noticeably augments the accuracy and completeness of the annotations, thereby heightening the model's ability to recognize lesion characteristics.

The finely corrected data is then fed back into the model for subsequent training iterations, aimed at strengthening the model's predictive capability and lessening reliance on expert intervention. On this basis, the precision and efficiency of the semi-automatic annotation process are dramatically improved through such iterative optimization, which is ultimately to heighten the model's ability to autonomously learn features from breast cancer images [33].

Table 2
Experimental Setup.

Device	Parameter
CPU	Intel Xeon E5-2678 v3
GPU	Eight NVIDIA RTX 3090
Memory	32 GB DDR4
Storage	1TB NVMe SSD

6. Experimental results

6.1. Data

The FFDM data employed in this study comprises two parts: (1) FFDM data from 1000 patients in the dataset provided by the First Affiliated Hospital of Zhejiang University; (2) Public data set VinDr-Mammo, which contains FFDM data from 5000 patients. Each patient's data includes both left and right breast images in the craniocaudal (CC) and mediolateral oblique (MLO) positions. The VinDr-Mammo dataset can be accessed at https://www.physionet.org/content/vindr-mammo/1.0.0.

6.2. Setup

During the training of Yolov7, a total of 19200 images from 4800 patients were used, comprising 800 cases from the private dataset and 4000 cases from the public dataset. The training-validation split was 8:2. Key experimental parameters were set as follows: batch size = 8, number of epochs = 100, initial learning rate = 0.01, with learning rate adjustment using cosine annealing.

In the training of HLSAM, the parameters of the pre-trained SAM model were fixed, and only the proposed HLSAM component was learned. The dataset consisted of the 19200 images used for Yolov7 training, and the model was trained on eight RTX 3090 GPUs with a total batch size of 32. The key parameters for the HLSAM experiment were: learning rate = 0.001, number of epochs = 12.

For performance evaluation of the Yolo-HLSAM model compared to other models, the remaining 800 cases from the private dataset and 4000 cases from the public dataset were used for validation. Details of the experimental setup are provided in Table 2 and all the programs were performed using codes developed in Python.

Table 3
Experimental results of attention mechanism comparison.

Attention mechanism			ACC%	TPR%	Dice%	mAP@ 0.5%	mAP@ 0.5:0.95%
CBAM	SE	ECA					
–	–	–	85.29	84.13	85.43	87.53	55.34
✓	–	–	**90.32**	**88.25**	**89.35**	**89.12**	**57.26**
–	✓	–	85.24	83.67	85.12	87.21	54.76
–	–	✓	84.32	83.71	84.61	86.91	54.62

6.3. Evaluation metrics

This paper is predominantly intended to accurately segment the segmentation of the relevant medical image data, so the true positive rate (TPR), false positive rate (FPR), precision (Pr), Dice index (DCS), Accuracy (ACC), and the mean average precision (mAP) at an IOU threshold of 0.5, denoted as mAP@0.5, and the mean average precision across IOU thresholds from 0.5 to 0.95, denoted as mAP@0.5:0.95, are employed as evaluation metrics. are used as the evaluation indicators.

6.4. Model performance evaluation

The performance evaluation of the recommended model is divided into two parts, that is, the evaluation of the region proposal network on the basis of Yolov7, and the performance evaluation of the HLSAM model. This section presents an analysis of the experimental results.

6.4.1. Performance evaluation of regional candidate networks based on Yolov7

To evaluate the effectiveness of the Convolutional Block Attention Module (CBAM) attention mechanism, a comparative analysis was conducted between the Yolov7 model and its three variants, each incorporating a different attention mechanism: CBAM, Squeeze-and-Excitation (SE), and Efficient Channel Attention (ECA). The models were assessed using key evaluation metrics, including accuracy, true positive rate (TPR), and Dice coefficient. The experimental results are summarized in Table 3.

The results presented in Table 3 indicate that, in comparison with the original Yolov7 algorithm, the incorporation of the CBAM attention mechanism enhanced the algorithm's accuracy by 5.03%, improved the true positive rate by 4.12%, elevated the Dice by 3.92%, the mAP@0.5 by 1.59%, and the mAP@0.5:0.95 by 1.92%. Conversely, integrating the SE attention mechanism gave rise to an abatement in accuracy by 0.05%, a reduction in true positive rate by 0.46%, and a decline in Dice by 0.31%, the mAP decreased by 0.32%, and the mAP@0.5:0.95 lowered by 0.58%. Similarly, adding the ECA attention mechanism gave rise to a drop of 0.97% in accuracy, a decrement of 0.42% in true positive rate, and an abatement of 0.82% in Dice score, the mAP@0.5 lessened by 0.62%, and the mAP@0.5:0.95 lowered by 0.72%. As these findings collectively suggest, both SE and ECA attention mechanisms not only diminished performance but also escalated computational demands on the model. In contrast, incorporating the CBAM attention mechanism yielded performance optimizations, which can be attributed to its dual reinforcement of channel-wise attention alongside spatial attention modules, thereby optimizing attention concentration across both dimensions. The accuracy of CBAM-Yolov7 and Yolov7 for test and training data is shown in Fig. 4.

In the context of FFDM images, the object detection process relies on the utilization of the Yolov7 network with CBAM attention mechanism. Evaluation indicators such as true positive rate, false positive rate, accuracy, Dice coefficient, and accuracy are employed. During the experiment, the performance was assessed across five confidence thresholds ranging from 0.1 to 0.5.

As evidently demonstrated by the research findings presented in Table 4, the network materializes a commendable true positive rate of 98.66% at a confidence threshold of 0.1. Nevertheless, the high rate of true positives is accompanied by a striking augment in the false positive rate, whereas the accuracy and mAP@ coefficient do not exhibit high values. Under such circumstance, it is imperative to weight the association between dissimilar metrics, so as to determine the optimal model parameter setting. It was finally decided to use a confidence threshold of 0.2. Despite the fact that the true positive rate lessens in this case, the false positive rate also diminishes substantially, which means that the model's predictions are more reliable. Simultaneously, the accuracy and mAP@ coefficient also reach the highest level separately. Such model performance betterment is crucial for accurate medical imaging diagnosis, especially in the analysis of digital breast medical photography (FFDM) data.

Table 4
Results of the confidence threshold comparison experiment.

Threshold	TPR%	FPR%	ACC%	Pr%	Dice%	mAP@ 0.5%	mAP@ 0.5:0.95%
0.1	**98.66**	50.14	65.93	49.14	63.52	58.62	29.26
0.2	91.35	13.31	**90.33**	**89.34**	**90.45**	**90.23**	**58.52**
0.3	63.17	10.86	81.83	74.32	68.12	67.56	34.62
0.4	53.76	7.63	80.66	75.40	63.53	62.34	29.68
0.5	29.43	**6.42**	72.47	69.22	41.39	45.52	21.67

In detail, a balance can be actualized between the true positive rate and the false positive rate by adjusting the confidence threshold, thereby obtaining more accurate and reliable results. This is not only crucial for heightening the performance of the current model but also provides valuable guidance for subsequent high-precision segmentation by adopting the SAM model. By refining model parameters, computer-aided diagnosis technology can be proficiently utilized to elevate the precision of early detection and diagnosis in breast diseases. This approach not only affords patients exceptional medical services but also bolsters their health and medical care support. Fig. 5 displays the detection effect of YOLOv7-CBAM algorithm on breast microcalcification lesion data set.

6.4.2. Performance evaluation of the Yolo-HLSAM model

The HLSAM model incorporates boundary-aware loss into HQ-SAM. To comprehensively assess the effectiveness of this modification, ablation experiments were conducted by comparing several critical performance metrics across dissimilar model configurations.

In accordance with Table 5, despite the fact that the initial index is low, with the introduction of boundary perception loss, each performance index shows a gradual betterment, especially in the multi-task setting, the boundary processing and small object detection accuracy of the model are substantially optimized.

The encoder utilized by HQ-SAM is available in three distinct sizes: Vit-b, Vit-l, and Vit-h, with Vit-b being the smallest model and Vit-h the largest model. With the purpose of determining the most suitable image encoder, the following experiments were performed, the results of which are revealed in Table 6.

As illustrated in the table above, a comparison across three different model sizes reveals that both HQ-SAM and Yolo-HLSAM demonstrate superior performance with Vit-h, achieving the highest TPR, ACC, and Dice scores. Specifically, Yolo-HLSAM achieves maximum values of 90.17 ± 0.89%, 90.63 ± 0.81%, and 89.95 ± 1.25%, respectively. This suggests that larger model sizes generally yield better refinement results in downstream tasks, indicating a positive correlation between

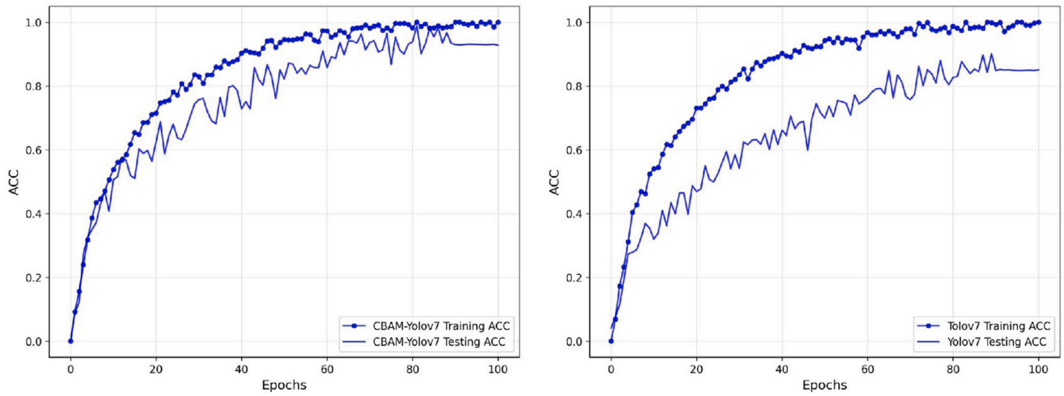

Fig. 4. CBAM-Yolov7 and Yolov7 Learning curve.

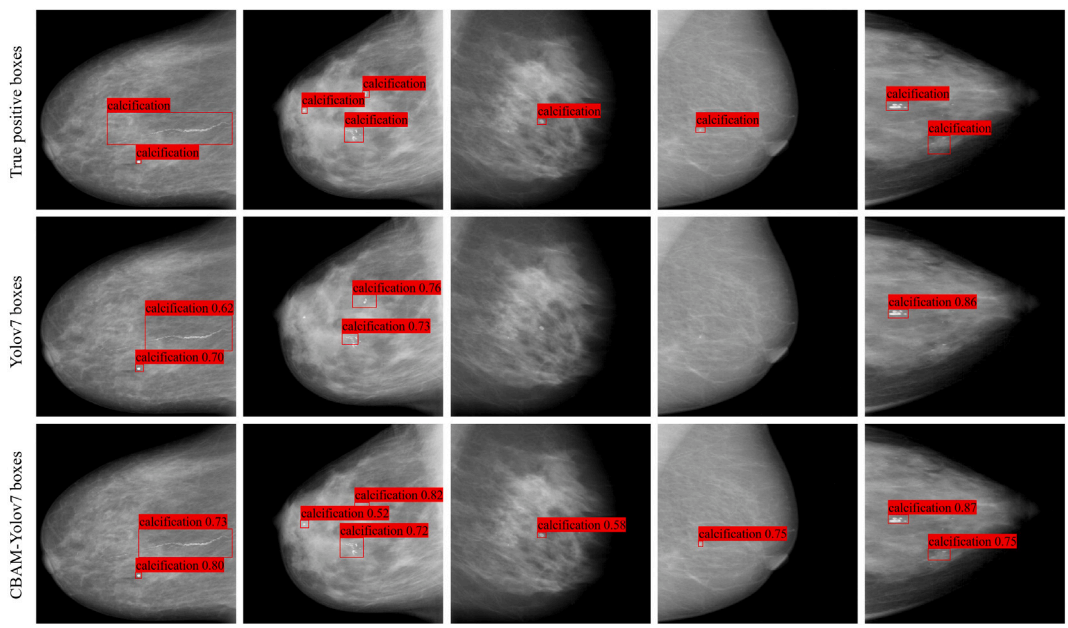

Fig. 5. CBAM-YOLOv7 network prediction result graph.

Table 5
Ablation of boundary-aware loss.

Model configuration	Global loss	Boundary-aware Loss	TPR%	ACC%	Pr%	Dice%
HQ-SAM	✓	–	74.82	68.42	61.53	67.22
HQ-SAM +Boundary-aware Loss	✓	Lightweight	75.24	68.73	61.79	67.89
HQ-SAM +Boundary-aware Loss	✓	Standard	75.68	69.86	62.74	68.34
HQ-SAM +Boundary-aware Loss	✓	Weighted	76.64	71.34	63.59	70.52
HQ-SAM +Boundary-aware Loss	✓	Multi-task	**80.32**	**76.43**	**69.52**	**75.24**

Table 6
Encoder type performance comparison experiment.

Encoder type	Model configuration	TPR%	FPR%	ACC%	Dice%
Vit-b		68.35 ± 1.52	29.14 ± 3.14	68.72 ± 1.83	67.65 ± 2.25
Vit-l	HQ-SAM	74.39 ± 2.26	23.87 ± 1.82	75.52 ± 2.08	75.35 ± 2.58
Vit-h		78.57 ± 1.85	20.45 ± 1.54	79.54 ± 1.72	78.32 ± 2.04
Vit-b		75.76 ± 2.21	22.42 ± 1.83	79.54 ± 1.72	78.32 ± 2.04
Vit-l	Yolo-HLSAM	82.45 ± 1.48	16.54 ± 1.25	83.42 ± 1.32	85.74 ± 1.82
Vit-h		**90.17 ± 0.89**	**9.89 ± 0.63**	**90.63 ± 0.81**	**89.95 ± 1.25**

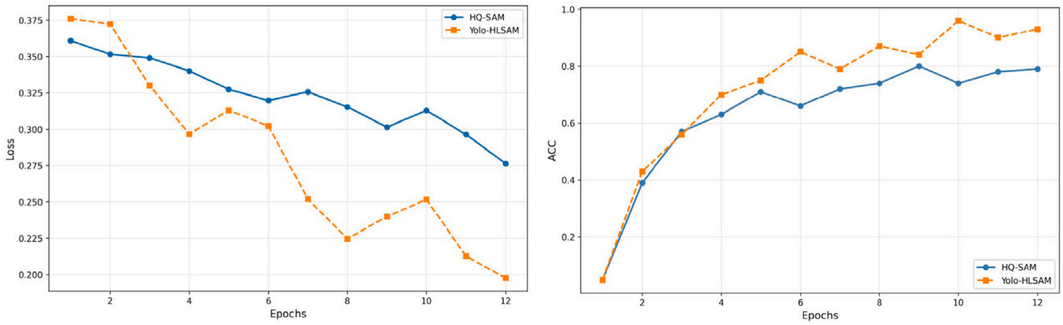

Fig. 6. The accuracy and loss curves for Yolo-HLSAM and HQ-SAM.

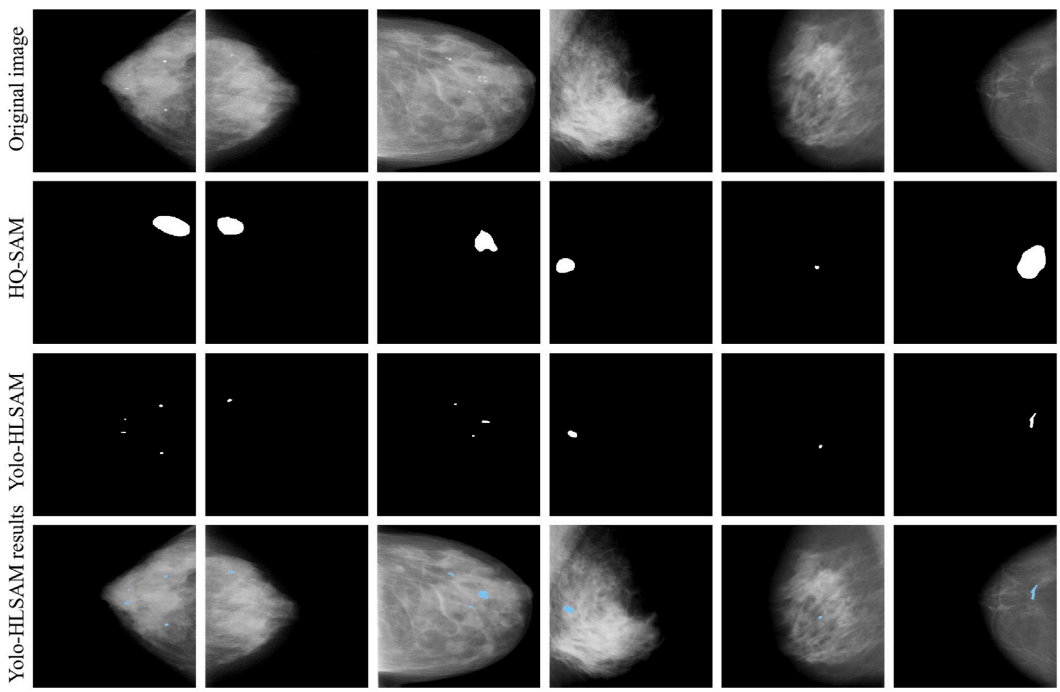

Fig. 7. Comparison between HQ-SAM model and Yolo-HLSAM.

Table 7
The comparison of HLSAM and HQ-SAM in training reasoning.

Method	GPU	Batch Size	Time(h)	FPS
HQ-SAM	8	32	4.07	4.8
HLSAM	8	32	4.11	4.7

model scale and processing capabilities. Larger models tend to possess a greater capacity for capturing and processing intricate details, which contributes to optimizing overall performance. Therefore, selecting a larger-scale model appears to be a viable strategy for improving result quality in high-precision applications. To further assess the impact of model optimization on performance, the loss and accuracy curves for Yolo-HLSAM and HQ-SAM are provided in Fig. 6.

The comparison of training performance between HLSAM and HQ-SAM is presented in Table 7. Both models utilize the ViT-L architecture. HLSAM outperforms HQ-SAM in segmentation while maintaining a similar training time.

In line with the results exhibited in Fig. 7, which illustrate the segmentation performance in four sample cases, there are more conspicuous differences in how the HQ-SAM and Yolo-HLSAM models cope with the segmentation of gland and lesion areas in FFDM images.

- Case 1:
  The HQ-SAM model incorrectly identified the glandular region as a lesion area, failing to accurately segment the true lesion. This misclassification highlights the HQ-SAM model's limitation in distinguishing between glandular tissue and actual lesions. On the contrary, the Yolo-HLSAM model was able to effectively differentiate between the glandular region and the lesion area, correctly identifying and segmenting the true lesion. This demonstrates Yolo-HLSAM's optimized ability to distinguish between different types of tissues.
- Cases 2, 3, and 4:
  In these cases, the HQ-SAM model was able to identify some of the lesion areas, but it failed to capture all the lesions accurately and therefore generated less detailed segmentations. On the contrary, the Yolo-HLSAM model not only successfully identified all lesion areas in these samples, but also provided more refined and detailed segmentation. This includes capturing the intricate boundaries of the lesions, which is crucial for precise medical analysis.

To thoroughly evaluate the performance of the Yolo-HLSAM model, we conducted a comprehensive comparison with several other models in the medical image segmentation task, including nn-UNet, Swin-UNet, DeepLab v3+, ResNet-50, SAM, Med-SAM, SAMed, and HQ-SAM. Both datasets were divided into 5 subsets, and the respective datasets were

Table 8
Model performance test results based on VinDr-Mammo dataset.

Model	ACC%	TPR%	Pr%	Dice%	MCC	Cohen's Kappa
nn-UNet	84.76 ± 1.23	86.21 ± 1.82	81.94 ± 2.15	82.56 ± 2.76	0.7638 ± 0.029	0.76 ± 0.03
Swin-UNet	88.64 ± 0.72	88.54 ± 1.35	88.67 ± 1.12	88.10 ± 1.85	0.8144 ± 0.024	0.81 ± 0.02
DeepLab v3+	82.01 ± 1.26	80.53 ± 2.31	82.54 ± 2.84	81.52 ± 3.17	0.6824 ± 0.035	0.68 ± 0.04
Resnet-50	79.28 ± 1.85	80.52 ± 2.51	78.96 ± 2.72	79.73 ± 3.18	0.5726 ± 0.045	0.57 ± 0.05
SAM	70.10 ± 3.85	65.12 ± 5.96	76.18 ± 3.05	72.53 ± 4.68	0.4681 ± 0.071	0.46 ± 0.08
SAMed	82.32 ± 1.23	81.25 ± 2.26	80.56 ± 2.61	80.91 ± 3.34	0.7357 ± 0.034	0.73 ± 0.03
Med-SAM	85.00 ± 1.09	87.31 ± 1.52	83.14 ± 2.91	85.18 ± 2.51	0.7481 ± 0.031	0.74 ± 0.03
HQ-SAM	80.29 ± 1.75	79.24 ± 2.85	80.95 ± 2.65	80.09 ± 3.15	0.6103 ± 0.037	0.61 ± 0.04
Yolo-HLSAM	**89.84 ± 0.57**	**90.05 ± 0.51**	**88.34 ± 0.67**	**89.19 ± 0.56**	**0.8144 ± 0.015**	**0.81 ± 0.02**

Table 9
Model performance test results based on Private data set.

Model	ACC%	TPR%	Pr%	Dice%	MCC	Cohen's Kappa
nn-UNet	83.74 ± 1.32	83.19 ± 2.02	80.85 ± 2.67	81.03 ± 3.23	0.6957 ± 0.027	0.69 ± 0.04
Swin-UNet	86.64 ± 0.95	86.46 ± 1.64	82.68 ± 2.71	87.34 ± 1.96	0.8131 ± 0.026	0.81 ± 0.02
DeepLab v3+	82.00 ± 1.26	83.63 ± 2.03	80.62 ± 2.59	81.50 ± 3.15	0.6148 ± 0.035	0.61 ± 0.04
Resnet-50	78.43 ± 2.13	77.85 ± 3.21	78.24 ± 2.92	78.05 ± 3.64	0.5848 ± 0.042	0.58 ± 0.05
SAM	66.19 ± 4.81	64.31 ± 6.17	63.21 ± 5.34	63.75 ± 6.82	0.3813 ± 0.083	0.38 ± 0.09
SAMed	82.04 ± 1.35	85.76 ± 2.04	74.86 ± 3.52	79.57 ± 3.23	0.6218 ± 0.032	0.62 ± 0.04
Med-SAM	83.79 ± 1.32	82.64 ± 2.35	79.92 ± 2.63	83.52 ± 2.64	0.7638 ± 0.028	0.76 ± 0.03
HQ-SAM	79.94 ± 1.95	82.29 ± 2.34	78.24 ± 2.96	78.27 ± 3.85	0.5642 ± 0.052	0.56 ± 0.05
Yolo-HLSAM	**89.57 ± 0.46**	**89.48 ± 0.31**	**88.31 ± 0.53**	**88.76 ± 0.32**	**0.8348 ± 0.022**	**0.83 ± 0.02**

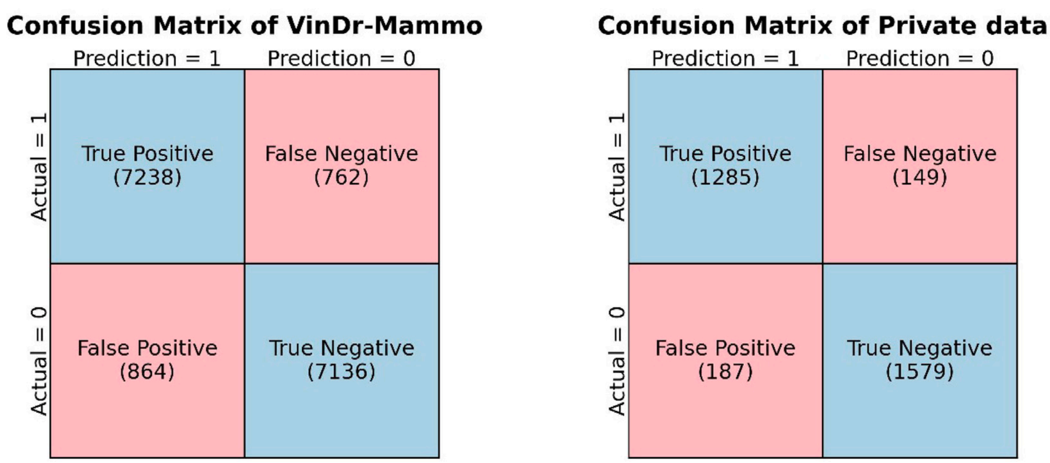

Fig. 8. Yolo-HLSAM's confusion matrix for these two data sets.

independently assessed using a 5-fold cross-validation method. The detailed results of the comparative study are presented in Tables 8 and 9.

As presented in Tables 4, 8, and 9, the Yolov7 method, with a low detection threshold, achieved a Dice coefficient of 90.45% and an accuracy rate of 90.33%, along with an exceptionally low FPR of just 13.31%. This demonstrates that this model is capable of attaining high accuracy while minimizing missed detections. Each experiment 5 times and report the mean and standard deviation. Specifically, Yolo-HLSAM performed the best, its achieved peak Dice coefficients of 89.19 ± 0.56% and 88.76 ± 0.32%, maximum accuracy values of 89.84 ± 0.57% and 89.57 ± 0.46%.

Further insights into the performance of Yolo-HLSAM in medical image segmentation are provided by the confusion matrices for the VinDr-Mammo and private datasets (Fig. 8). These matrices show that Yolo-HLSAM effectively maintains a reasonable TPR while controlling the FPR, which is essential for its robustness and suitability in complex medical image segmentation tasks. When compared to the SOTA model, Swin-Unet, Yolo-HLSAM demonstrated substantial improvements in the Dice coefficient, along with notable enhancements across other performance metrics. These results underscore Yolo-HLSAM's exceptional robustness and its promising potential for practical use in complex medical image segmentation scenarios.

6.4.3. Semi-automatic annotation based on Yolo-HLSAM

To present the entire process of YOLO-HLSAM more intuitively, the lesion area detected by YOLOv7 is marked with boxes. Then, each segment is shown one by one, and a distinct color is assigned to each of them. Finally, a visual composite diagram encompassing all the segmentation results is used to display the annotation effect more clearly. As shown in Fig. 9.

6.4.4. Limitations analysis of Yolo-HLSAM model

Although Yolo-HLSAM improves the performance of image segmentation, it still has limitations, particularly when handling small and complex regions. For instance, the high brightness of calcified areas can closely resemble the surrounding glands, leading to confusion and thus impacting the segmentation accuracy of YOLOv7. As shown in Fig. 10, despite YOLOv7's inability to effectively detect the lesion, resulting in inaccurate segmentation, the HLSAM model successfully delineates the lesion area once prompt points are provided. Consequently, future research should focus on enhancing the YOLO model's recognition accuracy in complex regions, particularly its capability to differentiate between high-brightness, similar areas. By incorporating more contextual information and employing advanced fine feature extraction strategies, the overall segmentation performance can be significantly improved.

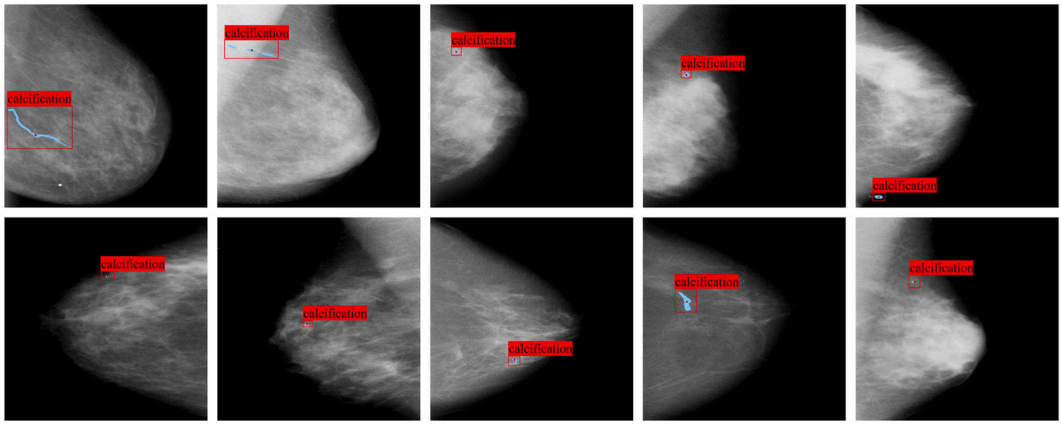

Fig. 9. Semi-automatic annotation based on Yolo-HLSAM.

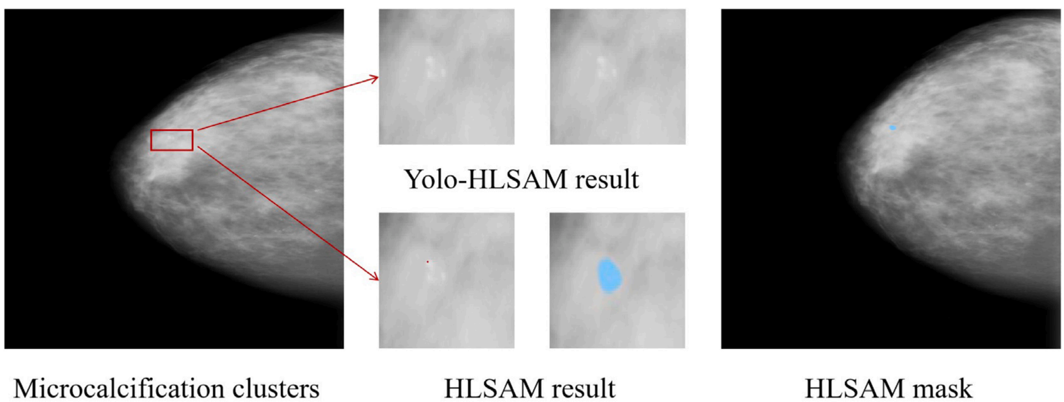

Fig. 10. Limitations of Yolo-HLSAM.

7. Conclusion

Medical image segmentation plays a paramount role in medical imaging, enabling doctors to accurately identify disease areas and guide treatment planning. In this study, we propose a framework for breast cancer microcalcification cluster segmentation based on Yolo-HLSAM, which leverages SAM technology to automate medical image segmentation, offering a revolutionary solution for medical image analysis. Additionally, the accuracy and efficiency of medical image segmentation are enhanced through the improvement and optimization of existing technologies, offering strong support for clinical research and practice. The core innovation of this framework lies in the Yolo-HLSAM model, which integrates unique technical features. Using the bounding boxes generated by the Yolov7 algorithm, the model extracts key point cues, which are then input into the HLSAM model to achieve precise segmentation. Notably, the incorporation of semi-automatic labeling technology significantly reduces the cost of manual labeling and improves work efficiency. Furthermore, an interactive mechanism is designed to allow users to select and modify segmentation results based on their needs, further improving the accuracy and reliability of the segmentation.

Despite the improvements in results, certain challenges and limitations remain. Future work will focus on two key areas for further enhancement and optimization. First, integrating pixel contrast learning among different classes is essential to improve the model's performance when handling challenging classes, particularly in complex-background and low-contrast image scenarios. Additionally, by thoroughly analyzing medical image characteristics and distributions, more effective feature extraction and classification algorithms will be developed. Incorporating contextual information is expected to improve segmentation accuracy, and combining this with the latest SAM-based medical applications [34] can further enhance the model's performance and applicability.

Beyond these, future research can explore deeper in the following potential areas:

1. Multi-modal Data Fusion: Currently, Yolo-HLSAM is mainly for single-modal image segmentation. However, medical images often involve multiple modalities (e.g., CT, MRI, ultrasound), each offering unique information. Future joint learning of multi-modal data can extract complementary insights to strengthen the model's segmentation in complex situations.

2. Self-supervised & Unsupervised Learning: Acquiring annotated medical image data is extremely challenging and costly. Thus, exploring self-supervised and unsupervised learning methods in the future can unearth latent patterns in unlabeled data to improve model performance.

3. Real-time Application & Computational Efficiency Optimization: Although current methods have increased segmentation accuracy, computational efficiency remains vital in real-time medical applications, like diagnostic systems. Future work should focus on minimizing model computational complexity and optimizing inference speed, perhaps via lightweight network structures, quantization, or knowledge distillation. This allows the model to boost efficiency without sacrificing accuracy.

Through continuous refinement, this method will undeniably play a more prominent role in medical image analysis, offer stronger support for clinical research and practice, and drive the further advancement of intelligent healthcare.

CRediT authorship contribution statement

Banteng Liu: Writing – review & editing. **Hongguang Chen:** Writing – review & editing, Writing – original draft. **Tianyi Zhu:** Supervision. **Zanting Ye:** Supervision, Conceptualization. **Haidong Cui:** Supervision. **Ke Wang:** Supervision.

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Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

I have shared the link to my code in the manuscript.

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